

Jhonier Negrete Muñoz

AI Transformation Lead | Enterprise AI Agents & Governance

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PROFESSIONAL SUMMARY

Forward-thinking AI Engineer specializing in organizational transformation through process agentification. Proven track record in designing autonomous agent architectures, orchestration platforms, and LLM governance layers that move AI from experimental demos to mission-critical production environments.

Expert in driving operational efficiency via RAG systems, Knowledge Graphs, and robust Eval frameworks. AI-driven mindset focused on delivering outsized impact and scaling digital interactions across Latam.

PROFESSIONAL EXPERIENCE

AI Engineer — GBM Colombia

May 2024 – Present · Medellín, CO

Strategic AI Transformation & Agent Architecture.

- **Process Agentification:** Architected and deployed enterprise-grade autonomous agent topologies using **LangChain** and **LangGraph**, incorporating complex delegation logic, escalation paths, and human-in-the-loop checkpoints to transform manual workflows into automated, auditable systems.
- **AI Governance & LLMOps:** Built a production-ready AI governance stack using **Bifrost** and **LiteLLM** for model proxying, routing, and policy enforcement, ensuring security and observability across multiple model providers.
- **Internal Orchestration Platform:** Engineered the company's internal AI orchestration platform to host agentic workflows, significantly accelerating the deployment of new AI initiatives.
- **Evals & Observability:** Implemented rigorous tracing and quality monitoring workflows using **Langfuse** and **Phoenix (Arize)** to ensure LLM reliability and performance consistency.
- **AI Transformation Proposals:** Lead the technical architecture and rollout planning for enterprise client engagements, aligning AI capabilities with high-level business KPIs.
- **Stack:** Python, FastAPI, LangChain, LangGraph, Bifrost, LiteLLM, Langfuse, Phoenix by Arize.

Junior Engineering Analyst — INTECOL SAS

Nov 2022 – Mar 2024 · Medellín, CO

- AI-enhanced video-surveillance application with real-time multi-object tracking (YOLO) on on-prem GPU.
- 3D-volume estimation with a TOF camera and Python point-cloud manipulation.
- Project lifecycle ownership across mechanical, automation and software teams.

Backend Engineer — MEMBO INVENTOS

Jul 2022 – Oct 2022 · Medellín, CO

- OCR pipeline processing **10,000+ PDFs** indexed in Elasticsearch; PyTorch + OpenCV.
- Implemented ML components (PyTorch, OpenCV) and contributed to large-scale backend systems.

PERSONAL PROJECTS

Founder — FiniaERP & NextEraTech

2026 · Personal AI-native projects

- **FiniaERP.com:** multi-tenant SaaS ERP pilot built with FastAPI, SQLAlchemy, Supabase, Next.js and TypeScript, including multi-tenancy, RBAC, usage tracking, CSRF protection and rate limiting.
- **NextEraTech.top:** digitalization pilot for SMEs, focused on workflows, dashboards and data collection.

- Built solo using **Cursor, Claude Code and Codex** as paired coding agents — hands-on practice in the AI-native development style I also apply in enterprise work.

CAPABILITY STACK

Autonomous agents: LangChain, LangGraph, agent topology design, delegation / escalation / human-in-the-loop, structured handoff, MCP-aware architectures.

Evals & Observability: eval harness design, LLM quality monitoring, tracing, Langfuse, Phoenix by Arize.

AI governance / LLMops: Bifrost (LLM gateway), LiteLLM (model proxy / routing), policy enforcement, model observability, audit logging, model routing across providers, build-vs-buy reasoning.

LLM / NLP: RAG-at-scale, Knowledge Graphs, prompt engineering, HuggingFace, LLaMA, BERTopic, DSPy.

AI-assisted development: Cursor, Claude Code, Codex.

Backend & Data: Python, FastAPI, SQLAlchemy, PostgreSQL, Supabase, Apache Spark, Kafka, Airflow.

Frontend: Next.js, TypeScript, TanStack Query, Zustand.

Cloud & Infra: AWS (EC2, S3, Lambda), Azure, Docker, Coolify, n8n, Notion API, GitHub Actions.

CV / Classical ML: PyTorch, Keras, TensorFlow, Scikit-Learn, OpenCV, YOLO.

EDUCATION

- **Specialization in Artificial Intelligence** — EIA University · Feb 2024 – Dec 2024. Capstone: *TTSFlow*: scalable TTS using MMS-TTS, FastAPI + AWS Lambda + Streamlit + Docker on EC2/S3.
- **Data Engineer Bootcamp** — DATAPATH · Nov 2023 – Feb 2024.
- **Mechatronics Engineering** — EIA University · Feb 2017 – Jun 2022.

LANGUAGES

Spanish (native) · English (Professional Working Proficiency).